On the Out-of-Sample Predictability of Short-Horizon Cryptocurrency Returns

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Abstract

We ask whether daily cryptocurrency returns can be forecast out of sample well enough to survive transaction costs. Three assets, two horizons, and six forecasters are evaluated by purged and embargoed walk-forward cross-validation over 2020–2026, giving 18 machine-learning settings. Our main finding is about the test rather than the data. The comparison the return-predictability literature makes by default—a conditional model against a random-walk, that is zero-return, benchmark, corrected for nesting by Clark–West with a Newey–West bandwidth of $h-1$—does not have its nominal size here. Under a nested-estimation null built by resampling the return series, which retains the drift, the tails, the feature persistence, the sample length and the estimators while removing conditional predictability, that configuration rejects 14–22% of the time at a nominal 5%. The distortion has an identifiable source: the estimators fit an unpenalised intercept, so the zero forecast is not their slopes-zero restriction, and the adjusted differential $\hat{f}_t = 2(y_t - \hat{y}_t^b)(\hat{y}_t^m - \hat{y}_t^b)$ picks up a drift term that is nonzero whatever the features do. Against the recursively estimated mean instead, measured size is 5–6% at $h = 1$—including for early-stopped gradient boosting—and about 10% at $h = 7$. Substituting the correct benchmark does not remove the rejections but relocates them: the raw count moves from 5 of 18 to 7 of 18 and the rejecting set changes from BTC-concentrated to SOL-concentrated. Against calibrated size the observed counts are within what the null produces, so we report no reliable evidence of predictability in either direction, and we do not claim its absence: no minimum detectable effect is established. Two supporting results are corrected in the same direction. The only significant sign-timing results, $p \approx 0.001$ at $h = 7$, arise from treating overlapping 7-day labels as independent and are $p \approx 0.24$–0.27 on non-overlapping subsamples. And the two settings whose net Sharpe intervals exclude zero lose most of it under a one-bar execution delay ($0.91 \to 0.21$), while carrying $\beta = 0.76$ and 0.89 to buy-and-hold with alpha t-statistics of 1.07 and 1.15—performance that is market exposure, measured rather than asserted.

1 Introduction

Claims of accurate cryptocurrency price prediction are abundant and rarely survive scrutiny. The reported accuracy is usually traceable to one of four errors, each of which is well documented and each of which is easy to commit without noticing.

Target leakage. The predicted quantity is a function of contemporaneous or future information, so the model fits an identity rather than forecasting [1]. Leakage of this kind is now understood to be a leading cause of irreproducible findings in machine-learning-based science more broadly [2].

Preprint. Code, data, and the full result tables: https://github.com/ITheClixs/crypto-return-predictability

Preprocessing leakage. A scaler, imputer, or feature selector is fit on the full sample before splitting, so the test distribution informs training.

Absent benchmarks. Accuracy is reported on a price series rather than on returns. Prices are close to a martingale, so a naive random-walk forecast attains near-perfect R^2; a model that appears to explain 99% of price variance may explain none of the return variance, which is the only component that can be traded [3].

Silent multiple testing. Many model, asset, and horizon combinations are tried and the best is reported without adjusting the threshold. In the closely analogous cross-sectional asset-pricing literature, Harvey et al. [4] argue that conventional significance levels are badly miscalibrated once the size of the collective search is acknowledged.

Preprocessing leakage is impossible by construction here, because standardisation is a pipeline step fit inside each fold. Target leakage is prevented by a test that perturbs future bars and asserts no feature at or before t changes. Multiple testing is reported over an explicit family, though not over the whole search. Benchmarking is the one item on the list this study initially got wrong, and correcting it is the paper's main contribution.

The standard tool for comparing forecast accuracy, the Diebold–Mariano test [5], assumes the competing forecasts are non-nested. The comparison the return-predictability literature actually makes—a conditional model against a random walk—violates that assumption, and Clark and West [6] give the correction. The applied question that then goes unasked is what the corrected test is testing. A regression fitted with an unpenalised intercept does not reduce to the zero-return forecast when its slopes are set to zero; it reduces to the training-window mean. Testing against zero therefore tests a joint hypothesis—no drift *and* no conditional predictability—while the sentence such tests are used to support is about the features alone. Where the drift is large, as it is for these assets over this window, the difference is not academic. Section 4.8 derives the contaminating term, Section 4.9 measures the resulting size distortion at 14–22% against a nominal 5%, and Section 5.2 shows that repairing the benchmark changes which settings reject rather than how many.

Our own first version of this study reported "five of 18 settings reject, against 0.9 expected from a search of that size." That comparison assumed the test was correctly sized. It is not, and the corrected expectation is 2.6–4.0. We report the earlier claim explicitly because the error is easy to make, is invisible in the code, and is not detected by any test of the kind we had written.

2 Related work

Return predictability and out-of-sample evaluation. Welch and Goyal [7] showed that a long list of equity-premium predictors that perform well in sample fail to beat a simple historical mean out of sample, and Campbell and Thompson [8] formalised the out-of-sample R^2 against that benchmark which we adopt in Section 4.6. The lesson generalises: in-sample fit is close to uninformative about forecast value, and the appropriate benchmark is a naive one estimated recursively rather than with hindsight.

Machine learning in asset pricing. Gu et al. [9] conduct a large-scale comparison of machine-learning methods for equity returns and find measurable but small out-of-sample gains, concentrated in methods that shrink aggressively. Our model roster is chosen in that spirit: regularised linear models and a deliberately constrained gradient booster, on the premise that an unregularised learner in this signal-to-noise regime fits noise and a null result obtained from an overfit model would be uninformative.

Backtest overfitting. Bailey et al. [10] and Arnott et al. [11] document how repeated backtesting manufactures apparently significant strategies, and Bailey and López de Prado [12] propose the deflated Sharpe ratio, which raises the bar in proportion to the number of trials. White [13] and Sullivan et al. [14] address the same problem for technical trading rules through a bootstrap reality check. We adopt the deflated Sharpe ratio together with explicit control of the false discovery rate and the family-wise error rate over our grid.

Cryptocurrency market efficiency. The empirical record is mixed and appears to be time-varying. Urquhart [15] reports evidence of inefficiency in early Bitcoin returns; Nadarajah and Chu [16] find

Table 1: Evaluated out-of-sample window per setting. Counts are forecasts, not trades; the trading rule of Section 4.5 samples every h-th bar.

Asset	h (days)	OOS forecasts	First	Last
BTC	1	2186	2020-07-23	2026-07-17
BTC	7	2174	2020-07-29	2026-07-11
ETH	1	2186	2020-07-23	2026-07-17
ETH	7	2174	2020-07-29	2026-07-11
SOL	1	1721	2021-10-31	2026-07-17
SOL	7	1709	2021-11-06	2026-07-11

that a simple power transformation of the same returns satisfies efficiency tests, and Tiwari et al. [17] report that Bitcoin is largely efficient over most of their sample. This literature evaluates statistical efficiency; it generally does not ask whether any detected dependence survives transaction costs, which is the question we take up.

Validation for dependent data. Purged and embargoed cross-validation is due to López de Prado [18], and is the mechanism by which we prevent a training label whose realisation overlaps a test window from informing the fit.

3 Data

We use daily adjusted OHLCV bars from Yahoo Finance for the USD pairs of BTC, ETH, and SOL. BTC and ETH span 2019-01-01 to 2026-07-18 (2,756 bars each); SOL begins 2020-04-10 (2,291 bars). Cryptocurrency markets trade every calendar day, so the bar index is contiguous with no weekend or holiday gaps. We verified this rather than assuming it, and Section 4.3 shows that the split logic remains conservative under gaps regardless.

The evaluated out-of-sample window is shorter than the data window because the first fold consumes 504 bars of training history. Table 1 gives the resulting per-setting counts.

Asset selection is not survivorship-free. BTC, ETH, and SOL were chosen in knowledge of their survival, while many contemporaneous tokens did not survive. This biases the study toward finding profitable long exposure, which strengthens rather than weakens a negative result about *predictability*, but would invalidate any cross-sectional claim. Section 7 returns to it.

4 Method

4.1 Features

Let C_t, H_t, L_t, V_t denote close, high, low, and volume at bar t. Every feature is a function of information available at or before t, and all twelve are scale-free so that a single model can pool across assets trading at very different price levels. We use cumulative log returns over $k \in \{1, 5, 10, 21\}$, realised volatility over $w \in \{10, 21\}$, Wilder's RSI over 14 bars recentred to roughly $[-1, 1]$, a price-normalised MACD histogram, two moving-average ratios, a mean normalised range, and a volume z-score:

$$r_t^{(k)} = \log \frac{C_t}{C_{t-k}}, \qquad \sigma_t^{(w)} = \mathrm{sd}\left(r_{t-w+1}^{(1)}, \ldots, r_t^{(1)}\right), \tag{1}$$

$$\mathrm{RSI}_t = 100 - \frac{100}{1 + \bar{U}_t/\bar{D}_t}, \qquad \mathrm{macd}_t = \frac{M_t - S_t}{C_t}, \tag{2}$$

$$\mathrm{ma}_t = \log \frac{\mathrm{SMA}_7(C)_t}{\mathrm{SMA}_{21}(C)_t}, \qquad \mathrm{dist}_t = \log \frac{C_t}{\mathrm{SMA}_{50}(C)_t}, \tag{3}$$

$$\mathrm{range}_t = \frac{1}{14} \sum_{i=0}^{13} \frac{H_{t-i} - L_{t-i}}{C_{t-i}}, \qquad z_t = \frac{\log(1 + V_t) - \mu_t^{(21)}}{s_t^{(21)}}, \tag{4}$$

where $\bar{U}$ and $\bar{D}$ are exponentially smoothed gains and losses with $\alpha = 1/14$, and $M_t = \mathrm{EMA}_{12}(C)_t - \mathrm{EMA}_{26}(C)_t$ with $S_t = \mathrm{EMA}_9(M)_t$. The longest warm-up is 50 bars; rows with any undefined feature are dropped.

The absence of look-ahead is enforced by a test rather than by inspection: future bars are perturbed and every feature value at or before t is asserted unchanged. This catches an accidental centred window, a sign error in a shift, or a non-causal smoother, none of which is reliably visible when reading code.

4.2 Prediction target

The label attached to a decision made at t is the return realised over $(t, t+h]$:

$$y_t^{(h)} = \log \frac{C_{t+h}}{C_t}. \tag{5}$$

The final h rows have no label. We retain them with features and a missing target, because that is exactly the state a live forecaster occupies on the most recent bar, and we exclude them from fitting and scoring.

4.3 Purged, embargoed walk-forward validation

Ordinary k-fold cross-validation shuffles time and trains on the future. A plain chronological split is also insufficient: because $y_t^{(h)}$ resolves at $t+h$, a training row close to the test block carries a label whose realisation overlaps the test window. Following López de Prado [18], we purge the h rows before each test block and impose a further embargo of e bars. Writing $\mathcal{T}_k$ and $\mathcal{V}_k$ for the train and test index sets of fold k, every fold satisfies

$$\max(\mathcal{T}_k) + h + e < \min(\mathcal{V}_k). \tag{6}$$

Test blocks tile forward without overlap. We use an expanding window with $|\mathcal{T}| \geq 504$ bars, $|\mathcal{V}| = 63$ bars, and $e = 5$, giving 35 folds for BTC and ETH and 28 for SOL. Figure 1 shows the resulting layout, drawn from the splits the study actually runs.

Purging is applied by row position, whereas the leak it prevents is a calendar one. Position-based purging remains valid under missing bars, and conservatively so: gaps only stretch the timeline, so h positions always span at least h calendar days. We assert this on a synthetic index with a quarter of its bars removed.

Two further controls are worth naming because they are commonly missed. Standardisation is a pipeline step fit inside each fold, so it never sees test rows. And the gradient booster purges its own early-stopping holdout: without a gap, the last h rows of its fitting slice carry labels realised inside that holdout, so early stopping would be tuned on partly observed outcomes. We apply the same h-bar purge one level down, and test it by corrupting exactly those rows and asserting the fitted model is unchanged.

4.4 Forecasters

Three benchmarks and three machine-learning models are refit from scratch on every fold. The benchmarks are the random walk $\hat{y}_t = 0$, the training-window mean $\hat{y}_t = \bar{y}_{\mathcal{T}}$, and an AR(1) regression of $y^{(h)}$ on $r^{(1)}$. The machine-learning models are ridge ($\alpha = 1$), elastic net ($\alpha = 10^{-3}$, $\rho = 0.5$), and XGBoost [19] with depth 3, learning rate 0.03, subsampling 0.8, $\lambda = 1$, and early stopping on the purged temporal holdout described above. The linear objectives are

$$\hat{\beta}_{\text{ridge}} = \arg\min_{\beta} \|y - X\beta\|_2^2 + \alpha\|\beta\|_2^2, \qquad \hat{\beta}_{\text{EN}} = \arg\min_{\beta} \frac{1}{2n}\|y - X\beta\|_2^2 + \alpha\rho\|\beta\|_1 + \frac{\alpha(1-\rho)}{2}\|\beta\|_2^2. \tag{7}$$

Both linear objectives are fitted with an unpenalised intercept on standardised features, and the gradient booster estimates its own base score from the training labels. Setting the slopes to zero therefore returns the training-window mean of the target, not zero: measured on BTC at $h = 1$, the ridge and elastic-net intercepts equal $\bar{y}_{\mathcal{T}}$ to machine precision. The forecaster these models nest is the training-window mean, and the zero-return benchmark is the further restriction that the intercept is also zero. Section 4.8 shows what that distinction does to the test.

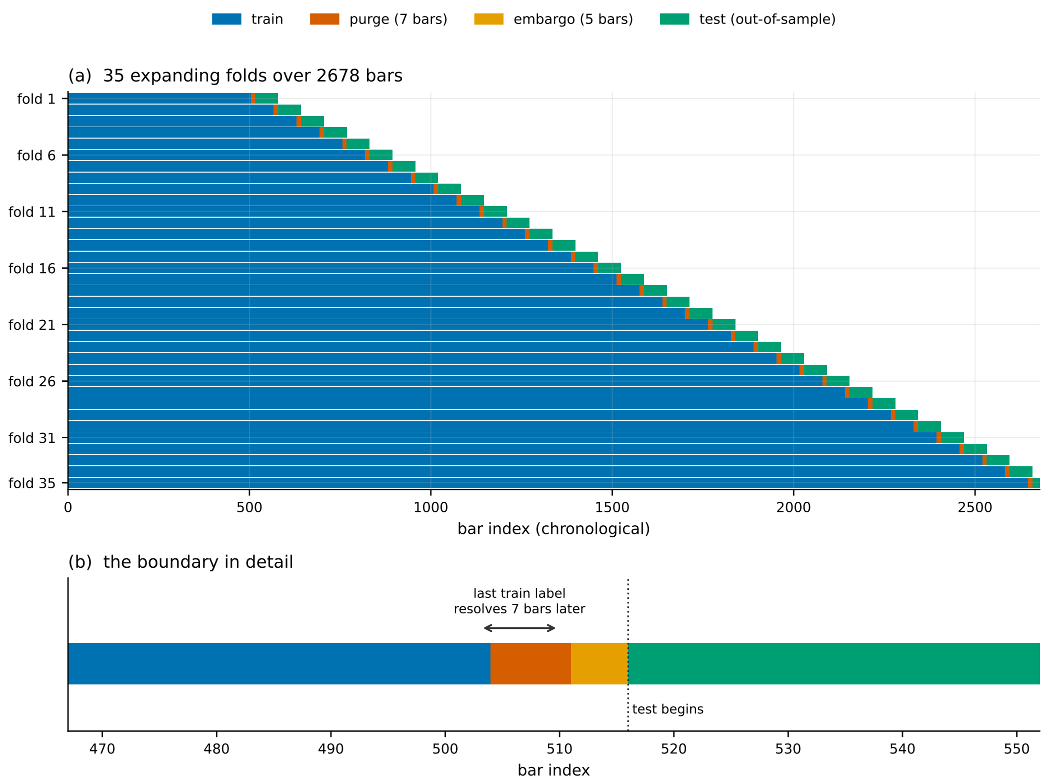

Figure 1: The validation design. Panel (a) shows all 35 folds on one timeline; panel (b) zooms on a single train/test boundary, where the purge and the embargo are individually visible. Equation (6) is what the gap enforces.

4.5 Trading rule and transaction costs

Forecasts become unit positions on a non-overlapping schedule: one decision every h bars, held to the next. Sampling every h-th bar prevents the same h-day return being counted h times. Costs are charged on turnover, so reversing a position pays for two sides:

$$w_t = \text{sign}(\hat{y}_t) \in \{-1, 0, +1\}, \qquad \tau_t = |w_t - w_{t-1}|, \qquad \pi_t = w_t\left(e^{y_t^{(h)}} - 1\right) - \tau_t c, \tag{8}$$

with $c = 17$ basis points per side (10 bp taker fee, 5 bp slippage, 2 bp half-spread). Equity compounds as $E_T = \prod_{t \le T}(1 + \pi_t)$ from unity.

Because decisions are taken every h bars, the Sharpe ratio annualises at $365/h$ rather than at 365. Using the latter inflates a seven-day strategy's Sharpe by a factor of $\sqrt{7} \approx 2.6$.

The schedule admits h start offsets and nothing distinguishes them. We therefore run every strategy on all h and report the spread, which Section 5.6 shows is not a formality.

An always-long reference is computed on the same schedule with the same cost model, so that a forecaster whose only achievement is holding the asset can be recognised as such.

4.6 Out-of-sample R^2_{OS}

We report the Campbell–Thompson form [8], whose denominator is the squared error of a genuine ex-ante benchmark forecast rather than of the realised mean of the evaluation window:

$$R^2_{OS} = 1 - \frac{\sum_t (y_t - \hat{y}_t)^2}{\sum_t (y_t - \hat{y}^b_t)^2}, \tag{9}$$

with $\hat{y}^{\,b}$ the recursively estimated historical mean. That benchmark scores exactly zero against itself, which serves as a wiring check on the metric. The alternative denominator $\sum_t (y_t - \bar{y})^2$ uses a quantity unknowable in advance and flatters the benchmark; we compute it as a cross-check but do not quote it.

Note that this is a *different* benchmark from the one the predictive-accuracy tests of Sections 4.7 and 4.8 use, and reporting effect sizes against one reference while testing significance against another is not defensible. It also happens to be the easier reference here: the zero forecast has lower squared error than the recursive mean in all six asset–horizon cells, by R^2_{OS} of $+0.0005$ (BTC h=1) up to $+0.1023$ (SOL h=7), the recursive drift estimate on SOL being badly wrong after the early-sample run-up. Measured against the zero forecast instead, exactly one of 18 settings has positive R^2_{OS}: BTC elastic net at $h = 1$, at $+0.0026$. Table 4 quotes the recursive-mean column; the zero-forecast comparison is the stricter one and is the number to read alongside a p-value computed against the zero forecast.

4.7 Diebold–Mariano and the nested-model problem

For loss differential $d_t = L(e_t^m) - L(e_t^b)$ under squared-error loss, the Diebold–Mariano statistic [5] is $\mathrm{DM} = \bar{d}/\sqrt{\hat{V}/n}$ with $\hat{V}$ a Newey–West long-run variance [20] truncated at $h-1$ Bartlett lags, appropriate because h-step forecast errors follow an MA($h-1$):

$$\hat{V} = \hat{\gamma}_0 + 2\sum_{k=1}^{h-1}\left(1 - \frac{k}{h}\right)\hat{\gamma}_k, \qquad \hat{\gamma}_k = \frac{1}{n}\sum_{t=k+1}^{n}(d_t - \bar{d})(d_{t-k} - \bar{d}). \tag{10}$$

The $1/n$ normalisation applies to every autocovariance including the lagged ones. This is deliberate: the Bartlett weights guarantee $\hat{V} \geq 0$ only under that normalisation, and the more intuitive $1/(n-k)$ can drive a weighted sum negative in small samples. We apply the Harvey–Leybourne–Newbold small-sample correction [21], $\mathrm{DM}^* = \mathrm{DM}\sqrt{(n + 1 - 2h + h(h-1)/n)/n}$, against a t_{n-1} reference. A negative statistic favours the model, and the test is two-sided because the null is equal accuracy and a model can be significantly worse.

The MA($h-1$) argument justifies the bandwidth for the *forecast error* under the null. It does not carry over to $\hat{f}_t$ once Equation (12) applies, because the drift component $2\mathbb{E}[y](\hat{y}_t^m - \hat{y}_t^b)$ inherits the persistence of the forecast. At $h = 1$ the bandwidth is zero lags, so no autocovariance enters at all, and four of the five settings that reject in Section 5.2 are at $h = 1$. We therefore report both $h-1$ and the Newey–West plug-in bandwidth $\lfloor 4(n/100)^{2/9} \rfloor$, which is 8 lags at $n \approx 2{,}200$. On this data the plug-in bandwidth yields *smaller* p-values—the adjusted differentials are mildly negatively autocorrelated, so the paper's choice is the conservative one at $h = 1$—but the count of surviving settings changes with it, and Table 2 shows that neither bandwidth is adequate at $h = 7$. A bandwidth cannot be chosen by the p-values it produces, so we report the sensitivity as a result rather than selecting.

4.8 Clark–West

Under a nested null, Diebold–Mariano is undersized as a test of predictability: the larger model pays sample mean squared prediction error to estimate coefficients that are truly zero, and the test reads that cost as evidence for the benchmark. Clark and West [6] subtract the term explicitly. With $\hat{y}^{\,b}$ the nested benchmark and $\hat{y}^{\,m}$ the larger model,

$$\hat{f}_t = \left(y_t - \hat{y}_t^b\right)^2 - \left[\left(y_t - \hat{y}_t^m\right)^2 - \left(\hat{y}_t^b - \hat{y}_t^m\right)^2\right] = 2\left(y_t - \hat{y}_t^b\right)\left(\hat{y}_t^m - \hat{y}_t^b\right), \tag{11}$$

and $\bar{f}/\sqrt{\hat{V}_f/n}$ is referred to a standard normal, one-sided. A *positive* statistic favours the model, opposite to Diebold–Mariano, which is reason enough never to report either p-value without its statistic.

What the benchmark choice does to the hypothesis. The right-hand identity in Equation (11) is the whole issue. Clark–West is a t-test of whether the model's *deviation from the benchmark* covaries positively with the outcome's deviation from it. With $\hat{y}^{\,b} = 0$ the identity collapses to $\hat{f}_t = 2y_t\hat{y}_t^m$, whose expectation is

$$\mathbb{E}\left[\hat{f}_t\right] = 2\left(\mathbb{E}[y]\,\mathbb{E}[\hat{y}^{\,m}] + \mathrm{Cov}(y, \hat{y}^{\,m})\right). \tag{12}$$

Table 2: Rejection rate at a nominal 5% under a nested-estimation null in which the features have no predictive value by construction. Monte Carlo standard errors are 1.2–2.5 percentage points for the 400-replication cells and 1.8–3.4 for the booster.

Statistic	Model	h=1 iid	h=1 block	h=7 iid	h=7 block
CW vs. zero forecast	ridge	14.2%	53.8%	23.5%	28.2%
CW vs. zero forecast	elastic net	22.0%	43.0%	26.2%	27.8%
CW vs. zero forecast	GBM	22.0%	—	—	—
CW vs. recursive mean	ridge	**6.2%**	42.8%	10.2%	11.2%
CW vs. recursive mean	elastic net	**6.2%**	28.2%	9.8%	10.8%
CW vs. recursive mean	GBM	**5.3%**	—	—	—
CW vs. zero, drift-only model	—	27.3%	27.0%	39.0%	34.0%
Diebold–Mariano vs. zero, two-sided	ridge	57.5%	15.2%	50.0%	46.5%

Only the second term is predictability. The first is drift multiplied by the average forecast, and it is nonzero whenever the sample has a drift and the model is on average long—which, through the fitted intercept, it always is here. BTC's realised drift over this sample is $+0.00102$ per day, 37% annualised. Measured across the 18 settings, that term accounts for 9% of the numerator at BTC $h = 1$ but 33% for the booster at $h = 1$ and 41–61% across BTC $h = 7$. The benchmark forecasters make the point most directly: the training-window mean, which uses no features at all, scores $\mathrm{CW} = +1.03$ against the zero forecast on BTC at $h = 1$.

We therefore report Clark–West against both the zero forecast, because that is what the applied literature does, and against the recursively estimated mean, which is the actual slopes-zero restriction of the estimators and the benchmark already used for R^2_{OS} in Section 4.6. Reporting one without the other conflates two hypotheses.

4.9 Calibrating the test instead of assuming it

Whether a rejection count is surprising depends on the test's size, which we measure rather than assume. Resampling the daily log-return series into a synthetic price path makes future returns independent of every past-information feature by construction, so the null "the features add no predictive value" is true, while the realised drift, the fat tails, the feature collinearity and persistence, the sample length, the purged walk-forward geometry, and the estimators themselves are all retained. Forecasts are genuinely estimated at every origin, so the estimation-noise term Clark–West exists to remove is present—which is what distinguishes this from a simulation in which the larger model's forecast is exogenous noise. In that latter construction the population mean squared prediction errors are σ_y^2 and $\sigma_y^2 + \sigma_u^2$, so the null of equal accuracy is false rather than true, no coefficient is estimated, and Clark–West centres for the algebraic reason that $\mathbb{E}[2y_t u_t] = 0$ for any independent u however poor a forecast it is. Such a design measures neither size nor bias, and an earlier version of this paper used it.

Table 2 gives rejection rates at a nominal 5% over 400 replications per cell (150 for the booster). The iid columns resample returns independently and so measure size. The block columns resample 21-day blocks, which preserves volatility clustering but can also carry genuine within-block dependence, so they bound the test's sensitivity to dependence rather than measuring its size.

Three readings follow. First, the configuration in standard use is oversized by a factor of three to four, and the drift-only row—27% rejection with no feature information whatever—locates the cause. Second, repairing the benchmark repairs most of the distortion at $h = 1$, and it does so for the early-stopped booster as readily as for ridge. That is a useful negative result about a live worry: greedy split selection, subsampling and a data-dependent number of trees do not visibly break the approximation here. It is calibration evidence for this estimator and this null, not a theorem. Third, the repair is incomplete. At $h = 7$ the corrected test still rejects about twice as often as nominal, which is the bandwidth question of Section 4.7; and under dependence-preserving resampling it is not trustworthy at any horizon, so a bootstrap on $\hat{f}_t$ rather than an analytic long-run variance is required for a definitive count. We do not have one, and we therefore do not report a definitive count.

4.10 Sign timing

A model can be poor at magnitude and useful at direction, which is all a sign strategy requires. Pesaran and Timmermann [22] test independence between the sign of the forecast and the sign of the outcome. With P the hit rate, P_y and P_x the proportions of positive outcomes and forecasts, and $P_* = P_y P_x + (1 - P_y)(1 - P_x)$,

$$S = \frac{P - P_*}{\sqrt{\widehat{\mathrm{var}}(P) - \widehat{\mathrm{var}}(P_*)}} \xrightarrow{d} \mathcal{N}(0,1), \qquad \widehat{\mathrm{var}}(P) = \frac{P_*(1-P_*)}{n}, \tag{13}$$

$$\widehat{\mathrm{var}}(P_*) = \frac{(2P_y - 1)^2 P_x(1-P_x)}{n} + \frac{(2P_x - 1)^2 P_y(1-P_y)}{n} + \frac{4P_yP_x(1-P_y)(1-P_x)}{n^2}. \tag{14}$$

A positive statistic indicates sign-timing skill and a negative one indicates a reliably wrong-way forecast. The statistic is undefined when the forecast never changes sign, which is precisely the case for the always-long drift models, and we report it as missing rather than as a number.

4.11 Sharpe ratio, deflation, and interval estimates

Sharpe ratios are noisy in short samples [23] and inflated by selection. The probabilistic Sharpe ratio [12] gives the probability that the true Sharpe exceeds a benchmark SR^*, correcting for sample length, skewness γ_3, and kurtosis γ_4:

$$\widehat{\mathrm{PSR}}(\mathrm{SR}^*) = \Phi\left[\frac{(\widehat{\mathrm{SR}} - \mathrm{SR}^*)\sqrt{n-1}}{\sqrt{1 - \gamma_3\widehat{\mathrm{SR}} + \frac{\gamma_4 - 1}{4}\widehat{\mathrm{SR}}^2}}\right]. \tag{15}$$

The deflated Sharpe ratio sets SR^* to the expected maximum across N trials, so the bar rises with the size of the search:

$$\mathbb{E}\left[\max_{i \le N} \mathrm{SR}_i\right] \approx \sigma_{\mathrm{SR}}\left[(1-\gamma)\,\Phi^{-1}\big(1 - \tfrac{1}{N}\big) + \gamma\,\Phi^{-1}\big(1 - \tfrac{1}{Ne}\big)\right], \tag{16}$$

with γ the Euler–Mascheroni constant. Here N is the number of models raced within one asset–horizon, which is a floor on the true search; the reported deflated Sharpe is therefore an upper bound and we label it as such. Confidence intervals come from a circular block bootstrap [24] with block length $\lfloor n^{1/3} \rceil$ and 500 resamples, reported as percentile intervals.

4.12 Multiple testing

Eighteen settings tested by a correctly sized 5% test would produce $0.05 \times 18 = 0.9$ rejections from noise alone. The test against the zero forecast is not correctly sized: at the rates in Table 2 the expectation is $18 \times 0.142 \ldots 0.220 = 2.6$ to 4.0. Against the recursive mean, using the per-horizon sizes, it is $9 \times 0.062 + 9 \times 0.10 \approx 1.5$. These are the numbers a rejection count must be read against, and the nominal 0.9 is the wrong one. We report raw counts alongside two adjustments of the Clark–West p-values over the family of 18. Holm–Bonferroni [25] controls the family-wise error rate and Benjamini–Hochberg [26] the false discovery rate:

$$p_{(i)}^{\mathrm{Holm}} = \max_{j \le i} \min\big\{(m - j + 1)\,p_{(j)},\, 1\big\}, \qquad p_{(i)}^{\mathrm{BH}} = \min_{j \ge i} \min\Big\{\tfrac{m}{j}\,p_{(j)},\, 1\Big\}. \tag{17}$$

Benchmarks are excluded from the family: they are the null being tested against, not candidates in the search.

5 Results

5.1 Point accuracy

Out-of-sample R^2_{OS} is negative in 16 of 18 machine-learning settings, from -0.0908 (ETH, $h = 7$, ridge) to $+0.0031$ (BTC, $h = 1$, elastic net). Rank information coefficients lie between -0.080 and $+0.053$. Figure 2 gives the full surface and Table 4 the underlying numbers.

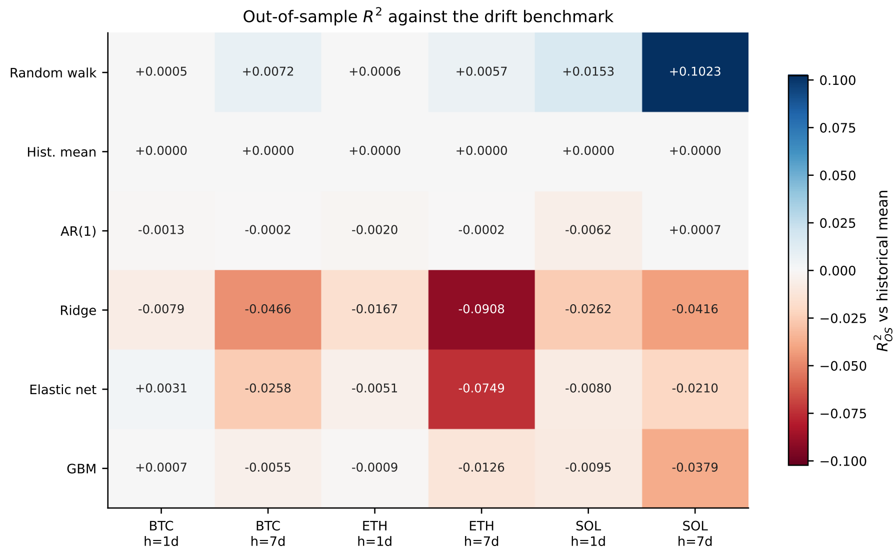

Figure 2: Out-of-sample R^2_{OS} against the drift benchmark. Red is worse than an ex-ante historical-mean forecast, blue better. The benchmark row is exactly zero by construction, which fixes the scale.

That ridge attains CW $= +2.62$ with $p = 0.004$ on BTC at $h = 1$ while its R^2_{OS} is -0.0079 has two explanations, and only one of them was in the earlier version of this paper. Clark–West does ask about *population* mean squared prediction error, and a model can have a lower population error while its own parameter-estimation noise leaves the realised forecast worse than a constant. But the two numbers are also computed against different benchmarks—the p-value against zero, the R^2_{OS} against the recursive mean—and, per Equation (12) and Table 2, the p-value comes from a test that rejects 14% of the time when nothing is predictable. The gap is not evidence of a weak signal. It is what an oversized test against a mismatched benchmark looks like.

5.2 Predictive accuracy against the random walk

Table 3 summarises. Under Diebold–Mariano the conclusion would be that machine learning is actively harmful here: no setting beats the random walk and seven lose to it significantly. Under Clark–West against the same zero benchmark, five settings reject at an uncorrected 5%. Both numbers derive from identical forecasts, and neither is the answer, because the second test rejects 14–22% of the time when nothing is predictable (Table 2).

Ordered by p-value the five are BTC ridge ($p = 0.004$), BTC elastic net (0.005), BTC GBM (0.022), and ETH ridge (0.031), all at $h = 1$, together with BTC elastic net at $h = 7$ (0.039). The sixth, BTC ridge at $h = 7$, misses at $p = 0.05004$; a count that turns on four parts in 10^5 should not be described as five.

The benchmark changes which settings reject, not how many. Substituting the recursively estimated mean—the actual slopes-zero restriction of these estimators—moves the raw count from 5 of 18 to 7 of 18 at the same bandwidth. It does not move the same settings. BTC ridge and elastic net at $h = 1$ survive ($p = 0.004$ both). BTC at $h = 7$ drops out ($p = 0.086$ and 0.061). And SOL, which shows no signal at all against the zero benchmark, becomes the strongest rejecter: SOL GBM at $h = 1$ goes from $p = 0.207$ to 0.031, SOL ridge at $h = 7$ from 0.923 to 0.022, SOL elastic net at $h = 7$ from 0.921 to 0.011. ETH at $h = 7$ moves further against the models.

The earlier reading of this table—four of five rejections in BTC, consistent with a weak short-horizon effect in the most liquid asset—was an artifact of the benchmark. Under the corrected benchmark the geography reverses, and under calibrated size neither pattern is distinguishable from noise. Multiplicity control does not settle it either: with the recursive-mean benchmark, 2 settings survive false-discovery control at the $h - 1$ bandwidth and 5 at the plug-in bandwidth, with 0 and 1 surviving family-wise control respectively. A conclusion that depends on the bandwidth is not a conclusion.

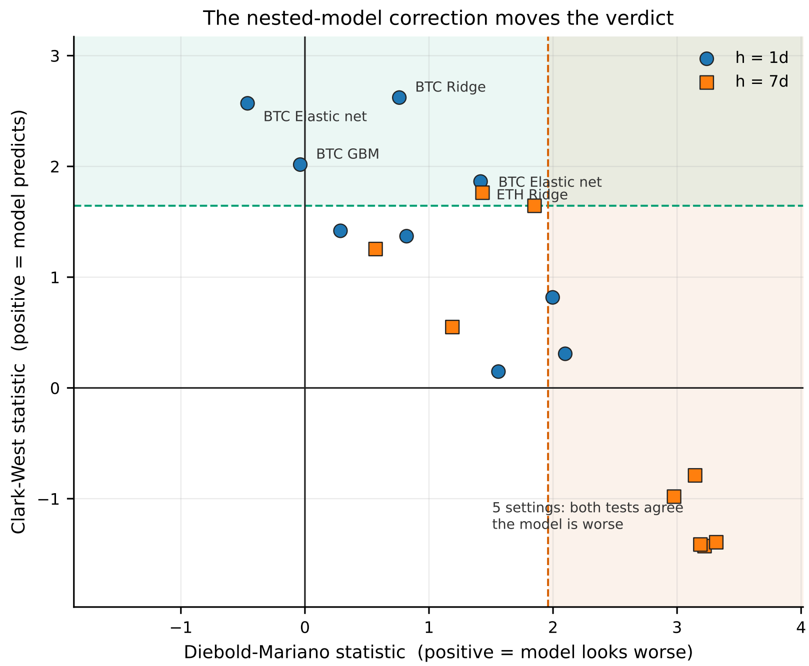

Figure 3: Every setting placed by its Diebold–Mariano statistic against its Clark–West statistic. The shaded vertical band marks where DM calls the model significantly worse; the horizontal band marks where Clark–West rejects no-predictability. Points falling in both are the settings DM misreads. The two tests disagree by construction under nesting, and the scatter shows the disagreement rather than asserting it.

Table 3: Outcome of each test and correction across the 18 machine-learning settings. Direction is required, not merely significance: a Diebold–Mariano rejection is attributed to whichever side the statistic's sign favours.

Criterion	Settings
Diebold–Mariano, favours model	0 / 18
Diebold–Mariano, favours benchmark	7 / 18
Clark–West, uncorrected	5 / 18
Clark–West, Benjamini–Hochberg	2 / 18
Clark–West, Holm–Bonferroni	0 / 18
Pesaran–Timmermann, sign skill	0 / 18
Pesaran–Timmermann, anti-predictive	2 / 18
Net Sharpe interval excludes zero	2 / 18
Expected by chance at $\alpha = 0.05$	0.9 / 18

5.3 Multiple testing

Controlling the false discovery rate at 5% leaves BTC ridge and BTC elastic net at $h = 1$, both with $p^{\mathrm{BH}} = 0.046$, marginally under the line. Controlling the family-wise error rate leaves nothing: the smallest raw p-value, 0.0044, does not clear the Holm threshold of $0.05/18 = 0.0028$ (Figure 4). Both corrections are applied to p-values from a test whose size is 14–22% rather than 5%, so they control the stated error rates only for a test that is correctly sized and here they do not. This is not a technicality that cuts one way: an oversized test inflates the raw p-values' evidential reading, while the multiplicity adjustment is calibrated to a nominal level the test does not have. The honest statement is that we cannot place a calibrated bound on the false-discovery rate over this family with the analytic test we have.

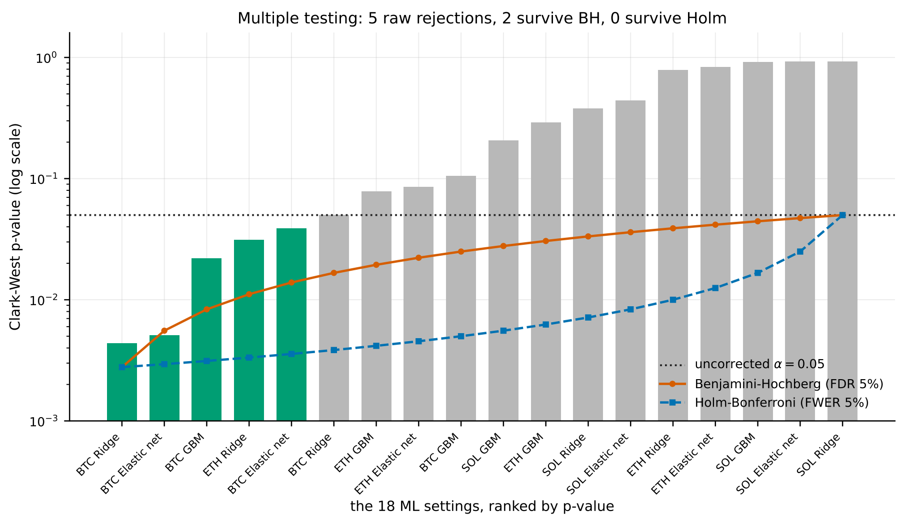

Figure 4: The 18 Clark–West p-values ranked against the Holm–Bonferroni and Benjamini–Hochberg thresholds, on a logarithmic scale because the entire decision occurs below $p = 0.05$. A bar must fall below a line to be rejected by that procedure. The figure's reference line is the nominal one; the calibrated expectation is 2.6–4.0 rejections against the zero benchmark rather than 0.9 (Table 2).

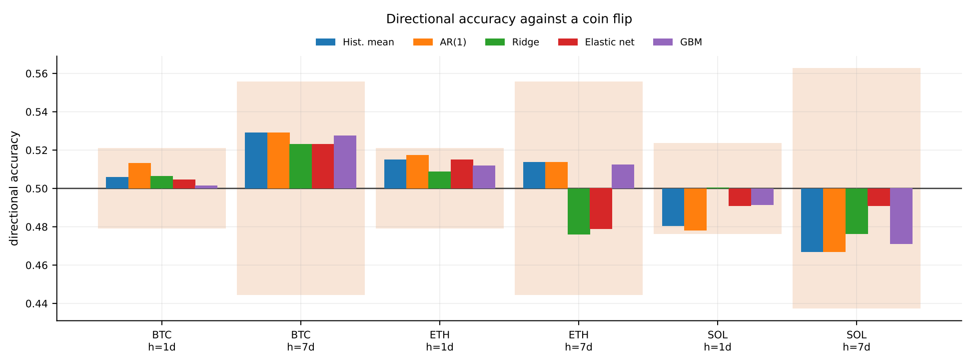

Figure 5: Directional accuracy against the band a fair coin would occupy. The band uses the effective sample size n/h: consecutive h-day forecasts are built from overlapping windows, and treating each as independent would shrink the band by $\sqrt{h}$ and manufacture significance.

5.4 Sign timing

Directional accuracy spans 0.471 to 0.528 and every setting lies inside its coin-flip band (Figure 5). No setting exhibits significant sign-timing skill.

The Pesaran–Timmermann variance terms all divide by n, so the statistic requires non-overlapping observations, and Table 4 reports it on all n rows of an h-step forecast whose labels overlap $h-1$ times. That inflates the statistic by roughly $\sqrt{h}$. It is the same error Figure 5 avoids by drawing its band at n/h, so the table and the figure disagree by a factor of $\sqrt{7}$ at the longer horizon. Recomputing on non-overlapping subsamples and averaging over all h phase offsets:

Setting	S (all n)	p	S (non-overlapping)	p
ETH ridge, $h = 7$	-3.29	0.001	-1.27	0.268
ETH elastic net, $h = 7$	-3.33	0.001	-1.27	0.243
SOL ridge, $h = 7$	-1.73	0.083	-0.66	0.597
BTC ridge, $h = 7$	$+1.41$	0.159	$+0.54$	0.592

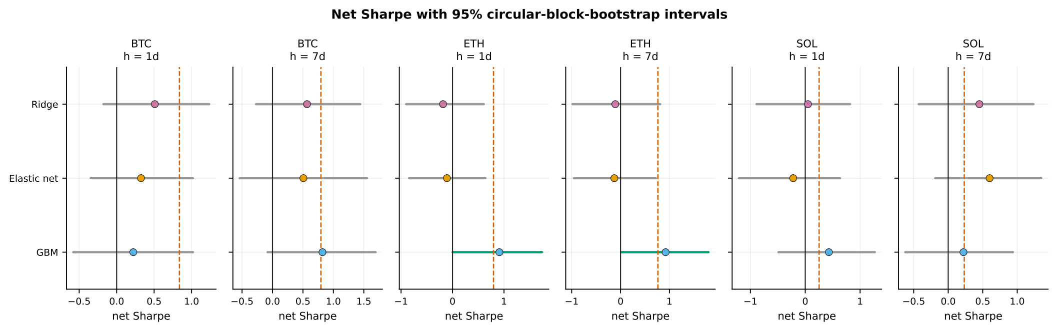

Figure 6: Net Sharpe with 95% circular-block-bootstrap intervals. The dashed line marks buy-and-hold on the same schedule and cost model. Intervals excluding zero are highlighted.

The two ETH settings were previously reported as significantly *anti*-predictive at $p \approx 0.001$. They are not significant at any conventional level once the labels are non-overlapping, and neither is anything else. The conclusion—no sign-timing skill in either direction—is unchanged; the evidence previously offered for its most striking part was invalid.

5.5 Economic performance

Two of 18 settings have a net Sharpe whose interval excludes zero, both gradient boosting on ETH: 0.91 with interval $[0.00, 1.74]$ at $h = 1$, and 0.92 with $[0.02, 1.80]$ at $h = 7$. Both intervals begin essentially at zero. Neither setting appears among the two surviving Benjamini–Hochberg correction, and neither of those two has an interval excluding zero.

The statistical winners and the economic winners are therefore disjoint. Two unrelated groups of survivors drawn from one search is the signature of noise: if a genuine short-horizon effect existed and were tradeable, the settings that detect it and the settings that profit from it ought to overlap.

Both economic winners are market exposure. Regressing each strategy's net return on buy-and-hold's, on the same schedule and cost model, with a Newey–West standard error on the intercept:

Setting	Sharpe	% long	β to buy-and-hold	α (ann.)	$t(\alpha)$
ETH GBM, $h = 1$	0.91	94%	0.76	$+0.236$	1.07
ETH GBM, $h = 7$	0.92	91%	0.89	$+0.192$	1.15
BTC GBM, $h = 7$	0.82	90%	0.82	$+0.097$	0.66
SOL elastic net, $h = 7$	0.60	56%	0.43	$+0.499$	1.23
Historical mean (all)	0.23–0.84	100%	1.00	≈ 0	$\leq \lvert 1.6 \rvert$

Neither of the two settings whose Sharpe interval excludes zero has an alpha distinguishable from zero once its market exposure is removed. This is the claim we previously made in words—that the high-Sharpe results are long-only exposure—and it survives being measured, which is the only reason it is worth stating.

Execution is assumed at a price the signal has not yet been computed from. Every feature at bar t uses the completed close C_t, and the backtest enters at that same C_t. Delaying entry by one bar, holding the forecasts fixed, moves the $h = 1$ results substantially and the $h = 7$ results hardly at all: ETH GBM at $h = 1$ falls from 0.91 to 0.21, BTC ridge at $h = 1$ from 0.51 to -0.23, ETH elastic net at $h = 1$ from -0.11 to -0.61; no $h = 7$ setting moves by more than 0.20. The one-day results are therefore properties of the execution convention as much as of the forecasts, and the surviving economic result—ETH GBM at $h = 7$, $0.92 \to 0.98$—is the one already shown above to be 91% long with $\beta = 0.89$.

Six of 18 settings out-Sharpe buy-and-hold on the point estimate, the weakest form of that claim given that every interval contains zero. Meanwhile the historical-mean forecaster attains Sharpe ratios of 0.84, 0.80, and 0.25 across the three assets while making exactly one position change. It

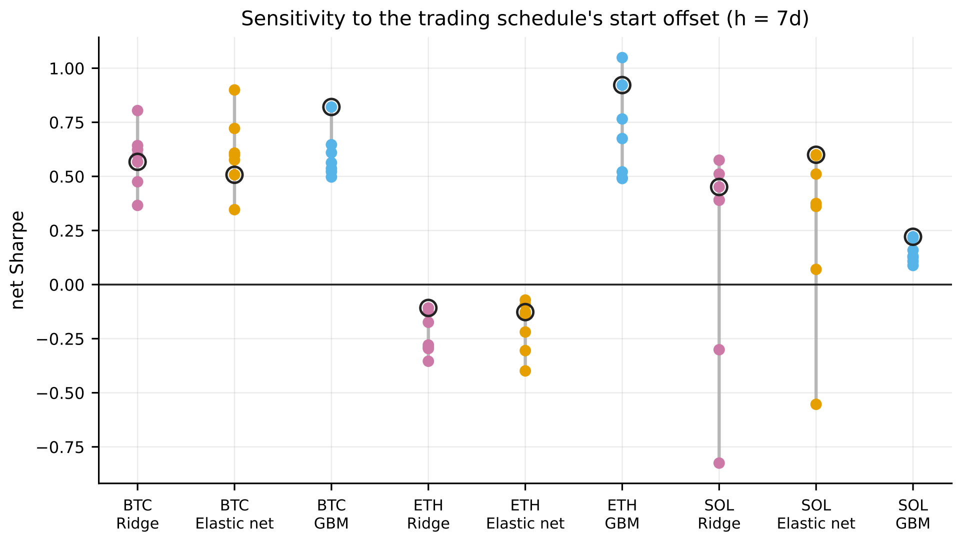

Figure 7: Net Sharpe on each of the h possible start offsets of the trading schedule at $h = 7$. The circled point is the offset the results table reports; nothing privileges it.

predicts a positive drift, hence is always long, and its net returns are numerically identical to buy-and-hold asset by asset and horizon by horizon. We carry buy-and-hold as an explicit row in Table 5 so that this cannot be mistaken for model performance.

Maximum drawdowns range from -55% to -99% across the machine-learning settings. This is a property of the position sizing rather than of any model: unit leverage reversing direction on roughly 70% annualised volatility carries a large variance drag, so the geometric return sits well below the arithmetic one. Buy-and-hold itself draws down between 76% and 96% over the same window.

5.6 Robustness to the trading schedule

Sampling every h-th bar leaves h schedules and nothing privileges the one starting at the first bar. At $h = 7$ the spread is substantial: SOL ridge reports a Sharpe of 0.45 drawn from a distribution running from -0.83 to 0.57, and SOL elastic net reports 0.60 from a range of $[-0.55, 0.60]$. The headline number is in part a property of where sampling happened to begin. A study reporting a single-offset Sharpe at a multi-day horizon without this check is reporting one draw and calling it an estimate.

6 Discussion

The benchmark defines the hypothesis, and the default one defines the wrong hypothesis. Diebold–Mariano is the default in applied forecast comparison and it is the wrong default whenever the benchmark is nested. But swapping in Clark–West is not sufficient, because the correction inherits whatever null the benchmark encodes. Against a zero-return benchmark, an estimator with a fitted intercept is tested for drift and conditional predictability jointly, and Equation (12) shows the drift enters the statistic directly. The measured consequence is a test that rejects three to four times too often, and a rejecting set whose membership changes completely—from BTC to SOL—when the benchmark is repaired. The lesson generalises past this dataset: a paper reporting Clark–West against a random walk has not stated its null until it has stated whether the larger model fits an intercept.

Calibrate the test; do not assume it. The distortion above is invisible in code review, survives a unit-test suite that checks every formula against hand-computed values, and is not detected by simulating a forecast that is independent noise—a design that measures neither size nor bias and that we used in an earlier version of this work. What detects it is a null built by resampling the data through the actual pipeline with actual estimated forecasts. That experiment costs a few minutes of compute and should be standard.

Most apparent performance was exposure, and it can be measured. The highest Sharpe ratios in the study belong to a forecaster that never changes its mind, and the two settings whose intervals exclude zero carry $\beta = 0.76$ and 0.89 to buy-and-hold with alpha t-statistics of 1.07 and 1.15. A one-bar execution delay removes most of the $h = 1$ performance. Three cheap controls—a buy-and-hold row, an exposure regression, and a delayed-entry variant—separate signal from exposure and from timing convention.

We do not read these results as establishing weak-form efficiency. An oversized test cannot bound predictability in either direction, and we report no minimum detectable effect, so this study is not powered to exclude an effect of economically relevant size. What it establishes is narrower and mostly methodological: on these three assets, at these horizons, with this feature set, no predictability claim survives a correctly benchmarked and size-calibrated test, and the claims that appeared to survive in an earlier reading were artifacts of the benchmark, the bandwidth, the label overlap, and the execution convention.

7 Limitations

Asset selection is not survivorship-free, as noted in Section 3. We use daily spot bars only, with no order book, funding rates, intraday structure, or cross-exchange information; short-horizon predictability, if it exists, is more likely to live at frequencies this data cannot resolve. The feature set is compact and the model families are three, so richer features or sequence models might change the picture, though the leakage discipline would not need to change with them. Short positions are modelled as fully collateralised, returning $-r$ with no borrow cost, margin requirement, or liquidation, which is reasonable for assessing signal quality and optimistic as an execution model. Costs are a flat per-side estimate that does not scale with size or volatility. Intervals are percentile bootstrap rather than BCa, adequate for a reject/do-not-reject reading and mildly biased for a statistic as asymmetric as the Sharpe ratio. The deflated Sharpe ratio understates the true search, deflating for the six models raced within one asset–horizon—three of which are benchmarks—rather than for the whole grid and for the design choices fixed before the grid was run, so it should not be read as a corrected number.

Four limitations bear directly on the results above rather than merely surrounding them. First, the size calibration in Section 4.9 resamples one asset's return series; size is not established for other assets, other frequencies, or nulls that preserve volatility clustering, where the corrected test rejects 28–43% of the time and no configuration we tested is usable. A definitive rejection count needs bootstrap inference on $\hat{f}_t$, which we do not have. Second, no minimum detectable effect is computed, so no statement in this paper bounds predictability from above. Third, the multiplicity family of 18 covers one statistic over one grid; the hyperparameters were fixed by hand and never tuned, so no conclusion here is about a model *family* rather than about $\mathrm{Ridge}(\alpha{=}1)$, $\mathrm{ElasticNet}(10^{-3}, 0.5)$, and one booster configuration. Fourth, the sample end date is unpinned: the default configuration resolves the end date to the current day, so a cache miss silently re-dates the study. The evaluated window is stamped into the generated report, and the committed data cache pins this paper's sample to 2026-07-18, but the pipeline as configured is not reproducible without it.

8 Conclusion

Across 18 model, asset, and horizon combinations evaluated by purged walk-forward cross-validation from 2020 to 2026, we find no predictability claim that survives a correctly benchmarked and size-calibrated test. The test in standard use—Clark–West against a zero-return benchmark at an $h - 1$ bandwidth—rejects 14–22% of the time on data constructed to contain no predictability, because a fitted intercept makes the zero forecast the wrong restriction and the sample drift enters the statistic directly. Repairing the benchmark brings measured size to 5–6% at $h = 1$, including for early-stopped gradient boosting, and relocates the rejections from BTC to SOL rather than removing them; against calibrated size, neither pattern is distinguishable from noise. The only significant sign-timing results were an overlapping-label artifact. The two settings with Sharpe intervals excluding zero carry $\beta = 0.76$ and 0.89 to buy-and-hold with insignificant alpha, and the one-day one loses three quarters of its Sharpe to a one-bar execution delay.

We do not claim that no cryptocurrency signal exists at daily frequency; this design is not powered to support that claim and we make no attempt to. The transferable result is that four routine choices—which benchmark, which bandwidth, whether labels overlap, and when execution is assumed—determined every apparent finding in an earlier version of this study, and that all four are cheap to check.

Reproducibility. All results, figures, and tables in this paper are generated by a single command from a public repository, and the tables reproduced here are emitted directly from the committed results file rather than transcribed. The test suite covers the leakage guarantees and checks the statistical formulas against hand-computed reference values. It did not, and could not, detect the benchmark misspecification of Section 4.8: every formula was implemented correctly and every test passed. The size calibration of Section 4.9, the corrected-benchmark re-test, the non-overlapping sign tests, the exposure regressions, and the delayed-execution backtest are separate scripts under `audit/`, each reproducing the numbers quoted for it, together with a full record of which earlier claims they overturn.

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A Full results

Table 4 gives forecast accuracy and all three predictive-accuracy tests for every model, asset, and horizon. Table 5 gives net-of-cost performance with bootstrap intervals, the phase spread, and both Sharpe corrections. Entries in bold are significant at the uncorrected 5% level; Section 5.2 gives the corrected counts, which are what should be read.

Table 4: Forecast accuracy and predictive-accuracy tests, all settings. Each test is reported as statistic (p). DM negative favours the model, CW positive favours the model, PT positive indicates sign-timing skill.

Model	RMSE	R^2_{OS}	Dir. acc.	Rank IC	DM (p)	CW (p)	PT (p)
BTC, $h = 1$							
Random walk	0.0301	0.0005	–	–	–	–	–
Historical mean	0.0301	0.0000	0.506	0.002	+0.18 (0.860)	+1.03 (0.152)	–
AR(1)	0.0301	-0.0013	0.513	0.033	+0.34 (0.737)	+1.24 (0.107)	+1.24 (0.214)
Ridge	0.0302	-0.0079	0.506	0.049	+0.76 (0.447)	**+2.62 (0.004)**	+0.51 (0.609)
Elastic net	0.0301	0.0031	0.505	0.046	-0.46 (0.644)	**+2.57 (0.005)**	+0.25 (0.806)
GBM	0.0301	0.0007	0.501	0.019	-0.04 (0.970)	**+2.02 (0.022)**	-0.38 (0.704)
BTC, $h = 7$							
Random walk	0.0804	0.0072	–	–	–	–	–
Historical mean	0.0807	0.0000	0.529	-0.018	+0.41 (0.685)	+1.01 (0.156)	–
AR(1)	0.0807	-0.0002	0.529	-0.023	+0.42 (0.675)	+1.00 (0.159)	–
Ridge	0.0825	-0.0466	0.523	0.044	+1.85 (0.064)	+1.64 (0.050)	+1.41 (0.159)
Elastic net	0.0817	-0.0258	0.523	0.043	+1.43 (0.152)	**+1.76 (0.039)**	+1.24 (0.217)
GBM	0.0809	-0.0055	0.528	0.032	+0.57 (0.569)	+1.25 (0.105)	+0.74 (0.459)
ETH, $h = 1$							
Random walk	0.0407	0.0006	–	–	–	–	–
Historical mean	0.0407	0.0000	0.515	0.019	+0.25 (0.806)	+0.81 (0.209)	–
AR(1)	0.0408	-0.0020	0.517	0.036	+0.47 (0.639)	+1.16 (0.124)	+0.96 (0.340)
Ridge	0.0411	-0.0167	0.509	0.053	+1.42 (0.157)	**+1.86 (0.031)**	+0.54 (0.588)
Elastic net	0.0409	-0.0051	0.515	0.040	+0.82 (0.413)	+1.37 (0.085)	+1.02 (0.308)
GBM	0.0408	-0.0009	0.512	0.021	+0.29 (0.775)	+1.42 (0.078)	-0.30 (0.766)
ETH, $h = 7$							
Random walk	0.1073	0.0057	–	–	–	–	–
Historical mean	0.1076	0.0000	0.514	0.039	+0.39 (0.697)	+0.89 (0.188)	–
AR(1)	0.1076	-0.0002	0.514	0.033	+0.40 (0.686)	+0.87 (0.191)	–
Ridge	0.1123	-0.0908	0.476	-0.047	**+3.15 (0.002)**	-0.79 (0.785)	**-3.29 (0.001)**
Elastic net	0.1115	-0.0749	0.479	-0.052	**+2.98 (0.003)**	-0.98 (0.837)	**-3.33 (0.001)**
GBM	0.1082	-0.0126	0.512	-0.042	+1.19 (0.235)	+0.55 (0.291)	+0.21 (0.835)
SOL, $h = 1$							
Random walk	0.0498	0.0153	–	–	–	–	–
Historical mean	0.0502	0.0000	0.481	-0.029	**+3.01 (0.003)**	-1.23 (0.891)	–
AR(1)	0.0504	-0.0062	0.478	-0.007	**+2.39 (0.017)**	-0.77 (0.779)	-0.81 (0.415)
Ridge	0.0509	-0.0262	0.500	0.010	**+2.10 (0.036)**	+0.31 (0.379)	+0.03 (0.977)
Elastic net	0.0504	-0.0080	0.491	0.013	+1.56 (0.119)	+0.15 (0.441)	-0.64 (0.522)
GBM	0.0505	-0.0095	0.492	0.023	**+2.00 (0.046)**	+0.82 (0.207)	+0.56 (0.576)
SOL, $h = 7$							
Random walk	0.1345	0.1023	–	–	–	–	–
Historical mean	0.1420	0.0000	0.467	-0.103	**+3.63 (0.000)**	-1.74 (0.959)	–
AR(1)	0.1419	0.0007	0.467	-0.098	**+3.62 (0.000)**	-1.71 (0.957)	–
Ridge	0.1449	-0.0416	0.476	-0.039	**+3.22 (0.001)**	-1.43 (0.923)	-1.73 (0.083)
Elastic net	0.1435	-0.0210	0.491	-0.034	**+3.19 (0.001)**	-1.41 (0.921)	-0.45 (0.649)
GBM	0.1446	-0.0379	0.471	-0.080	**+3.32 (0.001)**	-1.39 (0.918)	-0.49 (0.625)

Table 5: Net-of-cost performance. Sharpe ratios annualise at $365/h$. Phases gives the range of net Sharpe across all h start offsets of the trading schedule. DSR is an upper bound, as discussed in Section 7.

Model	Sharpe	95% CI	Phases	Max DD	PSR	DSR
BTC, $h = 1$						
Random walk	0.00	[0.00, 0.00]	–	0.0%	–	–
Historical mean	0.84	[0.01, 1.61]	–	-76.6%	0.98	0.87
AR(1)	0.21	[-0.51, 0.90]	–	-82.7%	0.70	0.34
Ridge	0.51	[-0.18, 1.23]	–	-65.7%	0.89	0.63
Elastic net	0.33	[-0.35, 1.02]	–	-74.3%	0.79	0.45
GBM	0.22	[-0.58, 1.02]	–	-89.4%	0.71	0.35
Buy & hold	0.84	[0.01, 1.61]	–	-76.6%	0.98	–
BTC, $h = 7$						
Random walk	0.00	[0.00, 0.00]	[0.00, 0.00]	0.0%	–	–
Historical mean	0.80	[-0.12, 1.63]	[0.78, 0.80]	-75.9%	0.98	0.83
AR(1)	0.80	[-0.12, 1.63]	[0.78, 0.80]	-75.9%	0.98	0.83
Ridge	0.57	[-0.27, 1.44]	[0.37, 0.80]	-78.5%	0.92	0.65
Elastic net	0.51	[-0.54, 1.55]	[0.35, 0.90]	-86.1%	0.90	0.60
GBM	0.82	[-0.08, 1.69]	[0.50, 0.82]	-54.6%	0.98	0.85
Buy & hold	0.80	[-0.12, 1.63]	–	-75.9%	0.98	–
ETH, $h = 1$						
Random walk	0.00	[0.00, 0.00]	–	0.0%	–	–
Historical mean	0.80	[-0.02, 1.57]	–	-79.4%	0.97	0.64
AR(1)	-0.15	[-0.96, 0.57]	–	-93.7%	0.36	0.02
Ridge	-0.18	[-0.90, 0.61]	–	-97.5%	0.33	0.02
Elastic net	-0.11	[-0.84, 0.64]	–	-97.7%	0.39	0.03
GBM	0.91	[0.00, 1.74]	–	-80.4%	0.99	0.74
Buy & hold	0.80	[-0.02, 1.57]	–	-79.4%	0.97	–
ETH, $h = 7$						
Random walk	0.00	[0.00, 0.00]	[0.00, 0.00]	0.0%	–	–
Historical mean	0.76	[-0.09, 1.56]	[0.71, 0.76]	-77.3%	0.97	0.62
AR(1)	0.76	[-0.09, 1.56]	[0.71, 0.76]	-77.3%	0.97	0.62
Ridge	-0.11	[-0.98, 0.81]	[-0.35, -0.11]	-99.2%	0.40	0.03
Elastic net	-0.13	[-0.95, 0.72]	[-0.40, -0.07]	-98.9%	0.38	0.03
GBM	0.92	[0.02, 1.80]	[0.49, 1.05]	-84.5%	0.99	0.76
Buy & hold	0.76	[-0.09, 1.56]	–	-77.3%	0.97	–
SOL, $h = 1$						
Random walk	0.00	[0.00, 0.00]	–	0.0%	–	–
Historical mean	0.25	[-0.63, 1.13]	–	-96.3%	0.71	0.43
AR(1)	-0.19	[-1.01, 0.67]	–	-97.5%	0.34	0.13
Ridge	0.05	[-0.89, 0.82]	–	-97.4%	0.54	0.27
Elastic net	-0.22	[-1.21, 0.63]	–	-97.1%	0.32	0.12
GBM	0.43	[-0.49, 1.27]	–	-91.3%	0.83	0.59
Buy & hold	0.25	[-0.63, 1.13]	–	-96.3%	0.71	–
SOL, $h = 7$						
Random walk	0.00	[0.00, 0.00]	[0.00, 0.00]	0.0%	–	–
Historical mean	0.23	[-0.75, 1.09]	[0.20, 0.25]	-96.2%	0.69	0.47
AR(1)	0.23	[-0.75, 1.09]	[0.20, 0.25]	-96.2%	0.69	0.47
Ridge	0.45	[-0.43, 1.23]	[-0.83, 0.57]	-89.7%	0.84	0.66
Elastic net	0.60	[-0.19, 1.35]	[-0.55, 0.60]	-83.9%	0.91	0.77
GBM	0.22	[-0.62, 0.94]	[0.09, 0.22]	-94.9%	0.69	0.46
Buy & hold	0.23	[-0.75, 1.09]	–	-96.2%	0.69	–

B Equity curves

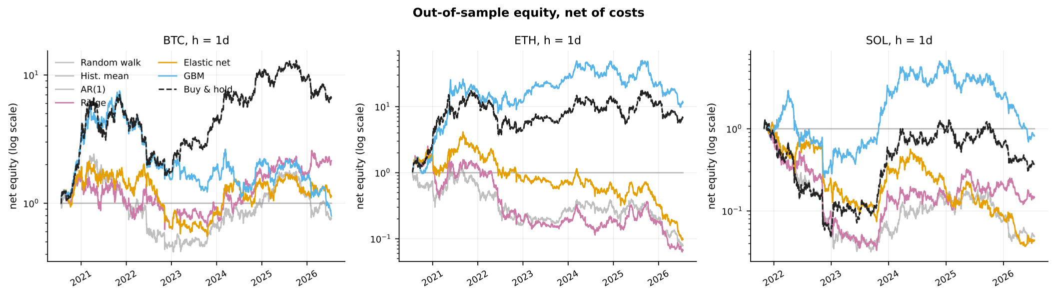

Figure 8: Out-of-sample equity net of costs at $h = 1$, against the always-long reference on the same schedule and cost model. Starting capital is unity and the vertical scale is logarithmic.